

VERRELL'S LAW

Retained-State Selection Framework

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"The past does not need to remain alive to keep affecting the future. Once a state has changed another state, that change can become part of the next selection environment."

Status of this document

This is the current public explanatory brief for Verrell's Law after the July 2026 retained-state hardening pass. It replaces earlier field-heavy public-download wording. Verrell's Law is presented here as a falsifiable retained-state selection framework. Collapse Aware AI is the engineering track. Field, electromagnetic, quantum-level, resonant, or broader physical interpretations are not ruled out, but they are not the load-bearing claim and are not presented as proven.

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Reader Orientation

This document explains Verrell's Law in its current public form: a retained-state selection framework. The core idea is that prior interactions can leave persistent state changes, and those retained changes can bias future selection.

The framework no longer asks a reader to accept a universal electromagnetic memory field, a consciousness-field model, or an Einstein-style tensor extension before the central claim can be evaluated. Those ideas may remain speculative research directions, but the current public framework stands on retained state.

Plain version

Things that happen to a system can leave a mark. If that mark persists, it can influence what the system does next. The future is not always selected from a neutral starting point; it can be selected from a system already weighted by its history.

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1. Current Position

Verrell's Law is currently presented as a falsifiable retained-state selection framework, not as established physics.

Core claim

Prior interactions can leave persistent state changes. Those retained changes can bias future selection, producing path-dependent divergence even when present inputs appear similar or unchanged.

The shortest spine remains:

Memory shapes weighting.
Weighting shapes selection.
Selection shapes what follows.

This does not abandon the wider vision. It puts the strongest, most testable part of the framework first. The deeper field, resonance, light, quantum-level, and physical-substrate questions remain open research directions rather than public proof claims.

Claim boundary

- Verrell's Law is not presented as established physics.
- It does not require memory to be stored in a universal electromagnetic field.
- It does not claim that consciousness has been solved.
- It does not claim that Collapse Aware AI proves new physics.
- It does not claim that softmax, Markov processes, Bayesian updating, reinforcement learning, hysteresis, or state-dependent selection are new inventions.

Discipline line

Fields may participate where measurable; retained state is primary.

2. The Retained-State Chain

The current public chain is:

Observation / Interaction -> State change -> Retained information -> Weighting / bias -> Altered future selection -> Path-dependent outcome

Observation does not mean mystical consciousness. It can mean measurement, input, contact, attention, state registration, environmental interaction, or computational update. The important point is that something enters the system and changes it.

Information becomes active when it is retained long enough to affect later weighting or selection. Memory is not important only because it is stored. Memory becomes active when it changes what happens next.

Bias means directional weighting

In ordinary language, bias often means error. In Verrell's Law, bias means directional weighting. A bias can stabilise continuity, preserve identity, support recognition, distort perception, create overfitting, or trap a system in an old loop.

Key distinction

Stored information is not automatically active. A vast archive can sit unused. A small retained trace can dominate selection if it carries enough weight.

3. What Is New Here - And What Is Not

A hostile reader may say: 'this is just hysteresis, a Markov process with state, an RNN, Bayesian updating, reinforcement learning, or ordinary memory.'

The correct answer is: the framework does not deny those precedents. The mathematics is deliberately standard because standard mathematics can be estimated, tested, compared, and falsified.

The contribution

The claim is not the novelty of the selection equation. The claim is that retained-state influence can be made explicit, measurable, governable, and testable through pre-committed baseline comparison.

Verrell's Law gives a name and a measurement discipline to a cross-system pattern: retained history can change the future selection landscape. The important move is not saying 'the past matters' in a vague way. The important move is asking how much retained state mattered, compared with a matched baseline.

What the framework keeps

- retained state as the primary mechanism;
- Active Information Weight as the measurable influence of retained prior information;
- Weighted Emergence Layering as the bridge between retained state and structured future influence;
- Collapse Aware AI as the engineering route;
- field activity as possible participant only where measurable.

4. The Mathematical Spine

The public mathematical spine treats retained memory/history as a bias term inside a standard selection model.

For candidate outcome y_i , define a candidate score or logit:

$$z_i = U(y_i; S_t, O_t) + \lambda B(y_i; M_t)$$

Selection probability is then:

$$P(y_i | S_t, O_t, M_t) = \exp(z_i) / \sum_j \exp(z_j)$$

Where:

Symbol	Meaning
S_t	current system state held matched by the test protocol
O_t	current observation, input, or interaction
M_t	retained memory, history, or state allowed to differ across conditions
y_i	candidate outcome, action, interpretation, or behaviour
U	baseline utility or present-state selection pressure
B	retained-state bias term
λ	retained-state influence coefficient

Equivalent baseline probability form

The same structure can also be written as:

$$P(y_i | S_t, O_t, M_t) = [P_0(y_i | S_t, O_t) \exp(\lambda W(M_t, y_i))] / [\sum_j P_0(y_j | S_t, O_t) \exp(\lambda W(M_t, y_j))]$$

These forms are equivalent when $U(y_i; S_t, O_t) = \log P_0(y_i | S_t, O_t)$, up to an additive constant, and $B = W$.

5. Baseline, Boundary, and Identifiability

A critic can fairly ask: if current system state already contains history, how do you separate present state from retained memory?

The answer is operational. The S_t / M_t boundary is defined by the test design, not by metaphysics.

Operational boundary

S_t means the state variables held matched across conditions by the protocol. M_t means the retained-state variables allowed to differ because the systems have different histories. What is controlled as present-state baseline belongs to S_t . What is tested as retained history belongs to M_t .

The baseline condition must be defined before testing. A valid baseline may be the same system with retained-state weighting disabled, neutralised, randomised, ablated, or otherwise held as absent for the tested regime.

Identifiability

Only the product of retained-state strength and retained-state bias enters the candidate score:

$$\lambda B(y_i; M_t)$$

So λ and B are not separately meaningful unless the baseline, retained-state feature, candidate set, and estimation rule are defined before testing. Without that discipline, any result could be fitted after the fact, which would weaken the framework.

Testing rule

Pre-commit the baseline, retained-history condition, candidate set, bias feature, estimator, confounds, and decision rule before looking at the result.

6. AIW, Estimator, and Falsification

Active Information Weight, or AIW, is the measurable influence of retained prior information on future selection probability.

A simple observable AIW effect is:

$$\text{AIW}(y_i) = P_M(y_i \mid S_t, O_t, M_t) - P_0(y_i \mid S_t, O_t)$$

The measurement question is not simply: what is stored? The sharper question is: what still has weight?

Estimator discipline

The public estimator uses log-odds comparison to reduce the softmax selection model to a linear relationship in the retained-state influence term. Across controlled trials, the retained-state coefficient can be estimated and tested against a null baseline.

Plain version: hold the present input stable, vary the retained history, and measure whether future selection changes in a repeatable direction beyond the baseline.

Falsification

- If matched systems with different retained histories do not diverge, the claim is not supported in that tested regime.
- If retained-state influence is statistically indistinguishable from the null baseline, the claim is not supported in that tested regime.
- If pre-enumerated confounds explain the divergence, the retained-state claim fails in that tested regime.
- If bias-off and bias-on conditions do not differ, the engineering analogue fails in that tested regime.

Correct null wording

The right failure statement is: 'refuted in the tested regime.' Not: 'disproven everywhere.' A failed test constrains the tested setup, proxy, model, and regime.

7. Collapse Means Selection Resolution

The word collapse can attract the wrong fight. In this public framework, collapse does not need to mean quantum wave-function collapse.

Definition

Collapse means the resolution of competing candidate outcomes into a selected outcome under present input, retained state, and governing constraints.

A selected outcome can be an action, interpretation, behaviour, prediction, classification, biological response, social consequence, or other resolved system state.

Memory is not passive retrieval

Verrell's Law does not deny biological memory, synaptic plasticity, neural traces, engrams, distributed neural networks, cellular change, embodied state, or environmental cue loops.

The extension is that recall and behaviour may involve reconstruction under weighting. A memory can return changed. A place can reactivate an old frame. A sound, smell, image, or conversation can pull a system into an earlier selection pattern. These are not proofs of new physics. They are human-facing examples of retained state becoming active again.

8. Field and Physical-Extension Boundary

Verrell's Law does not currently identify a final storage substrate for retained informational influence.

It does not rule out field-based, electromagnetic, quantum-level, photonic, resonant, biological, environmental, or broader informational mechanisms. It simply does not require them as proven assumptions.

Disciplined field position

Fields may participate; retained state is primary. If field activity contributes to retained-state influence, that contribution must be measurable, unit-defined, instrument-defined, noise-controlled, and predictive beyond ordinary biological, environmental, or computational variables.

No general field-coupling measurement is claimed in this document.

Physical extension

The current public framework does not modify general relativity, quantum mechanics, thermodynamics, neuroscience, or standard probability theory. Whether structured retained information has any physical stress-energy contribution is an open, currently unformulated question.

That is why the physical-extension branch is treated as speculative archive material, not the current canonical form of the law.

9. Collapse Aware AI Boundary

Collapse Aware AI is the engineering implementation track. It is not the full proof of Verrell's Law. It is not a quantum computer. It is not a claim that current AI systems are conscious. It is not ordinary memory storage.

CAAI public definition

Collapse Aware AI is proprietary middleware for governed retained-state behavioural selection. It governs how retained state is allowed to influence final behaviour without retraining the underlying model.

A base model, game engine, agent framework, or simulation system may generate candidate behaviours. CAAI sits around that process and governs how retained state influences which behaviour is selected.

Public runtime shape

Host system / engine / model -> structured input payload -> candidate behaviours
-> retained-state influence -> Governor constraints -> selected behaviour ->
diagnostic evidence / fallback status -> persistent state update

The commercial question

Does retained state influence future behavioural selection in a controlled, inspectable, useful way compared with a baseline?

Boundary

CAAI proves or fails as engineering. Verrell's Law proves or fails as research. They are related, but not identical.

10. What the Framework Claims - and Does Not Claim

What it claims

- systems are not always neutral at the point of selection;
- retained history can matter;
- memory-derived weighting can bias what comes next;
- this influence can be measured through divergence between matched systems with different retained histories;
- the past can remain active when it changes the state from which the future is selected.

What it does not claim

- completed experimental proof;
- a solved theory of consciousness;
- memory exists only outside the brain;
- AI systems are conscious;
- CAAI proves new physics;
- a universal electromagnetic memory field is required;
- standard softmax, Bayesian updating, reinforcement learning, control theory, hysteresis, or path dependence are new inventions.

Narrow public claim

Retained informational history can be treated as an active, measurable weighting influence on future selection, producing path-dependent divergence under controlled conditions.

11. One-Page Structural Summary

Layer	Meaning	Status
Retained state	Prior interaction leaves persistent change.	Core framework
Active information	Stored information becomes active when it changes selection.	Measurable principle
Bias	Directional weighting, not automatically error.	Definition
Selection	Possible outcomes resolve into an expressed outcome.	Core process
AIW	Measured probability shift caused by retained prior information.	Measurement layer
WEL	Layering of retained influence into structured future weighting.	Bridge concept
Fields / resonance	Possible contributors only where measurable.	Open research
CAAI	Governed retained-state behavioural selection in software.	Engineering track
Falsification	Same input, different history, measured divergence or no divergence.	Scientific discipline

Final shape

Verrell's Law is best understood as a retained-state selection framework. It does not say the past magically controls the future. It says something sharper: the past can remain active when it changes the state from which the future is selected.

Collapse Aware AI tests one narrow engineering form of this. The broader framework keeps open the deeper question of whether retained informational influence also appears through biological reconstruction, environmental coupling, fields, resonance, light, frequency, or physical substrate effects.

Closing note

This document is intended as a strong public explanatory brief, not a claim of completed physics. It preserves the open field question while keeping the public position disciplined: retained-state influence is the core, CAAI is the engineering route, and deeper substrate claims remain open until tested.

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